Designing CIC filter :-

1] The parameters I decided upon were $N = 2$, $m = 1$, $R = 2048$

Reason for this selection

1] So basically the output from the moderator has the output as

$$y[n] = y[n-1] + e_q[n] - e_q[n-1]$$

so basically the error term was

$$y[n] - x[n] \approx e_q[n] - e_{q-n-1}$$

and this running sum of error function is bounded so not ideal choice would be go for more sampling more N would be better but issue with it is like summing again

Coz the change $x[n] - x[n-1]$

$x[n] = A\sin(wn)$ fun is $\pm 2A\sin w/2$

w is very small as $\frac{990}{409\,6000}$ fun sample so negligible

is basically we are summing up the previous sum like average & average in time domain is convolution with rectangular wave i.e multiplying by $(\text{sinc})^n$ in frequency domain

So droop is

$$|H(f)| = \left| \frac{\sin \pi R m f / f_s}{R m \sin \pi f / f_s} \right|^n$$

10 Hz – about 0 DB

500 Hz – about −0.91 dB

990 Hz – about −3.84 dB → 0.643 times amplitude

So a high drooping factor is added to signal

Also the issue is about band pass width increasing N improves attenuation around the output band CIC doesn't know which frequency to attenuate cause for 990. The maximum frequency can be represented without aliasing is

Nyquist $\frac{2000}{2} = 1000$ Hz

So if an unwanted signal above 1 KHz aliases & becomes 990 KHz so higher N attenuates this unwanted one also so we used N=2 not greater and N=1 didn't give good results

2] $m = 1$ → It placed the first null at highest frequency possible didn't do unnecessary narrowing of the main lobe and reduces additional attenuation

$f_{null} = \frac{f_s}{Rm}$

So Now $m = 1$

$f_{null} = 2$ KHz

at $m = 2$

$f_{null} = 1$ KHz → Kinda completely suppress

at say $m = 4$, or 3

990 is not in main lobe

Hogenauer's $= B_{in} + \lceil N \log_2 Rm \rceil$

$= 24$ bits

So made the Substraction registers of 24 bits

FIR Coefficients

1) Used cutoff as 1 KHz → as more f we are putting is 990 Hz and sampling at 2 kHz

2) Order is kept at 50 because, initially I tried with lower number as 25 also i had to keep the parameters

P = N = 2 → Vertical Bending → as more N more droop so have to be more aggressive upward curve

C = m × 0.5 = 0.5 → Horizontal Stretching → it is about null as first null at 2KHz at C = 0.5 We know our signal is in the first half part

So considering these parameters 26 ninjas didn't work so I increased the number to get desired results.

Things I changed in my testBench

Given Test Bench

```
end
always @(posedge clk) begin
    $fwrite(fd1, "%d\n",  bits);
end
always @(posedge clk) begin
    if(outready)begin
        $fwrite(fd2, "%d\n",  $signed(cicout));//keep 20 bits width
    end
end
always @(posedge clk2) begin
    $fwrite(fd3, "%d\n",  $signed(firout)); //keep 20 bits width
end
always @(posedge clk3) begin
    $fwrite(fd4, "%d\n",  $signed(cicout));//keep 20 bits width
end
```

10 MHz

```
always #50 clk=~clk;
always #25600 clk2=~clk2; //change according to filter output frequencies
always #102400 clk3=~clk3; //change according to filter output frequencies

integer i;
initial i = 0;

integer fd1,fd2,fd3,fd4;
initial begin
    fd1 = $fopen("bits.txt","w");
    fd2 = $fopen("cic512_990.txt","w"); //ouput file change according to need
    fd3 = $fopen("fircomp.txt","w"); //ouput file change according to need
    fd4 = $fopen("cic512_500_by4.txt","w"); //ouput file change according to need
end
```

19.9 KHz

4.8 KHz

R = 10 MHz / 4.8 KHz = 2048

My changes

```
always #50 clk=~clk;
always #25600 clk2=~clk2; //change according to filter output frequen
// Keep the same clk3 frequency, but align its first rising edge
// just after the first valid FIR output.
initial begin
    clk3 = 0;
    #205100;
    clk3 = 1;

    forever begin
        #102400 clk3 = 0;
        #102400 clk3 = 1;
    end
end
```

```
integer fd1,fd2,fd3,fd4;
initial begin
    fd1 = $fopen("bits.txt","w");
    fd2 = $fopen("cic2048_10.txt","w"); //c
    fd3 = $fopen("fircomp_10.txt","w"); //c
    fd4 = $fopen("validfir2048_10.txt","w");
end
```

2048

Clocks frequency is same

```
always @(posedge clk) begin
    $fwrite(fd1, "%d\n",  bits);
end
always @(posedge clk) begin
    if(outready)begin
        $fwrite(fd2, "%d\n",  $signed(cicout));//keep 20 bits width
    end
end
always @(posedge clk2) begin
    $fwrite(fd3, "%d\n",  $signed(firout)); //keep 20 bits width
end
always @(posedge clk3) begin
 if (outready2) begin
    $fwrite(fd4, "%d\n", $signed(firout));
 end
end
```

So the flow what i understood

Input clk

4096 + 26 samples loaded

fd2 : # ones acc outready

outready on at every 2048 cycles of main clock for decimation R = 2048

So we get about 2000 samples

fd3 : clk2 which is 512 times main clock we get 8000 samples here

fd4 : clk 3 which is 2048 we get 2000 samples

So when you name fd1 = cic 512

fd3 = cic 512 - by-4

according to me it is same coz both work at every 2048 clock cycles

and fd3 what we call fircomp is working every 512 cycles even when outready2 is not working so has repetitions coz it should also work at 2048 clock cycles

We did this so that output clk 3 samples the FIR output when outready 2 is high

Originally clk high at 102400, 307200, 512000

outready high at 20 5050, 409950, 614650...

because of #resist sins

I then renamed files because

fd2 - cic name at every 2048 clock cycle so cic2048

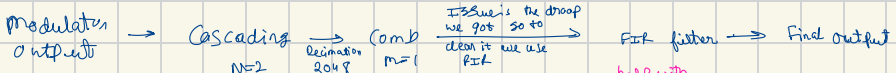
fd3 - fircomp → has the extra ones cause worked even when outready2 not high

fd4 - validfir2048 - coz worked at only every 2048 cycles and yeah it worked when outready2 high like according to me right output that we want.

also I added

$fclose(fi) i = 1, 2, 3, 4 at end coz down simulation then i had to put manually in console so fix it to write valid output

Architecture

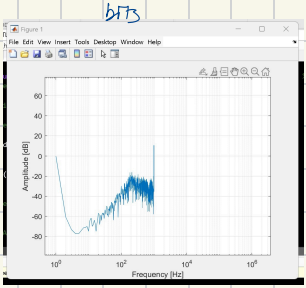


Results Next Page

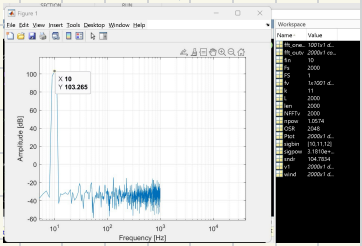
Results [The changes I did in Testbench affects FIR output CIC remain same]

10 Hz

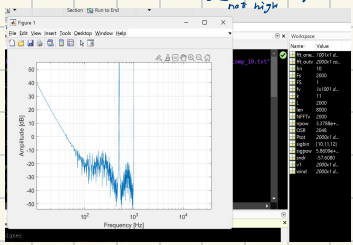
bits



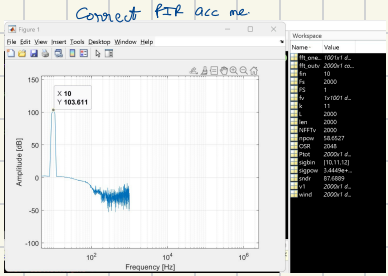
CIC output



Pin at per SIZ clock also has issues at outready2 not high

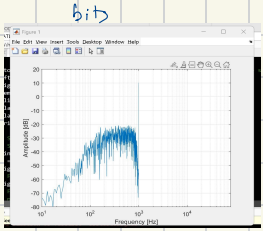


Correct FIR acc me

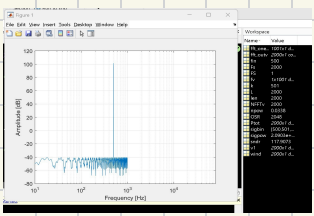


500 Hz :-

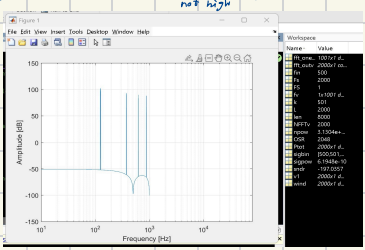
1) bits



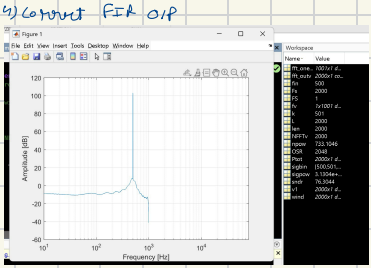
2) CIC



3) Pin at per SIZ clock also has issues at outready2 not high

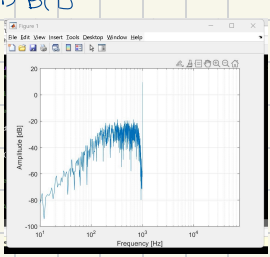


4) Correct FIR o/p

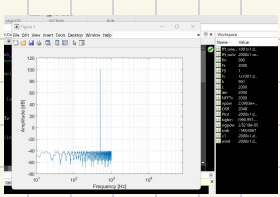


990 Hz

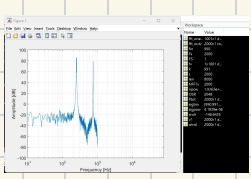
1) Bits



2) SNDR output



3) Pin at per SIZ clock also has issues at outready2



4) Correct FSR

